F78AB - Actuarial and Financial Mathematics B
Tutorial 4 – Stochastic interest rate models

Question 1

1. Let i_n denote the interest rate from time $n-1$ to time n. It is assumed that

$$i_n = \begin{cases} 0.03 & \text{with probability } 1/3 \\ 0.07 & \text{with probability } 2/3. \end{cases}$$

   Calculate the expected value, the variance and the standard deviation of $(1+i_n)$.

2. Let S_T be the accumulation after T years of £1 invested at time 0. Calculate $E[S_3]$ and $Var[S_3]$ on the assumption that

   (a) $i_1, i_2, \ldots$ are independent and identically distributed.

   (b) $i_1 = i_2 = \cdots$ with probability 1.

Question 2 Let i_t denote the effective rate of interest from time t to $t+1$ (note the time labelling!). It is assumed that for $t = 0, 1, 2, \ldots$

$$i_{t+1} = \begin{cases} i_t + 0.02 & \text{with probability } 0.25 \\ i_t & \text{with probability } 0.5 \\ i_t - 0.02 & \text{with probability } 0.25. \end{cases}$$

Given that $i_0 = 0.06$, calculate the probability that an investment of £1 at time 0 will accumulate to more than £1.20 at time 3.

Question 3 The rate of interest in any year has a mean of 6% and standard deviation of 1%. The yield in any year is independent of the yields in all previous years. Calculate the standard deviation of the accumulated value at time 12 of an investment of £10 at time 0.

Question 4 The random variable i_n denotes the effective rate of interest per annum from time $n-1$ to time n, for $n = 1, 2, \ldots$. Suppose that $(i_n)_{n=1,\ldots,\infty}$ are independent and identically distributed, with distribution:

$$i_n = \begin{cases} 0.03 & \text{with probability } 0.2 \\ 0.05 & \text{with probability } 0.6 \\ 0.07 & \text{with probability } 0.2. \end{cases}$$

The random variable A_n denotes the accumulation to time n of the following cashflows: £1 invested at time 0, £1 invested at time 1, …, and £1 invested at time $n-1$.

1. Calculate $E[1+i_n]$ and $E[(1+i_n)^2]$.

2. Calculate $E[A_1]$ and $E[A_2]$.

3. Calculate the variance of A_2.

Question 5 The effective rate of interest earned in the year from time $t-1$ to t is denoted by i_t. It is assumed that $i_1, i_2, \ldots$ are independent and identically distributed, with $\mathbb{E}[i_t] = 0.08$, $\mathbb{E}[i_t^2] = 0.042$ and $\mathbb{E}[i_t^3] = 0.013$.

Let A_n represent the accumulation up to time n of £1 payable at times $0, 1, \ldots, n-1$. Using a spreadsheet or other means to do the calculations,

1. Calculate $\mathbb{E}[A_{10}]$.
2. Use a recursive algorithm to calculate $\mathbb{E}[A_{10}^2]$.
3. Derive a recursive algorithm for calculating $\mathbb{E}[A_n^3]$, and evaluate $\mathbb{E}[A_{10}^3]$.

Question 6 The investment return earned in the year from time $t-1$ to t is denoted by i_t. Assume $(1+i_t)$ is lognormally distributed. The expected value of the investment return is 5%, and the standard deviation is 11%. Returns are assumed to be independent and identically distributed.

1. Calculate the parameters of the lognormal distribution of $(1+i_t)$.
2. Calculate the probability that the rate of interest in the year from time $t-1$ to t lies between 4% and 7%.
3. Suppose that £100 has been invested at time 0. Calculate the probability that this will accumulate to more than £120 at time 2.

Question 7 Let i_t be a random variable representing the interest rate in year t, measured from some origin. $\{i_t\}_{t=1}^{\infty}$ is a sequence of independent and identically distributed random variables. The random variable $(1+i_t)$ is lognormally distributed and has mean $\mathbb{E}[1+i_t] = 1.10$ and standard deviation $\mathrm{sd}(1+i_t) = 0.15$. The random variable v_t is defined as

$$v_t = (1+i_t)^{-1}.$$

1. Show that the mean and standard deviation of v_t are 0.925995 and 0.126271, respectively.
2. Calculate the mean and standard deviation of the present value at time 0 of the combined payment stream of £1 payable at time 2 and £1 payable at time 4.

Question 8 In any year, the yield on funds invested with a given insurance company has mean value j and standard deviation s, and it is independent of the yields in all previous years.

1. Derive formulae in terms of j and s for the mean and the variance of the accumulated value after n years of a single investment of 1 unit at time 0.
2. Let i_n be the rate of interest earned in the nth year. Each year the value of $(1+i_n)$ has a lognormal distribution with parameters $\mu = 0.08$ and $\sigma = 0.04$.

   Calculate the probability that a single investment of £1 000 will accumulate over 16 years to more than £4 250.

Question 9 Let X be a random variable which models business losses. Fix $\alpha \in (0,1)$. The *Value-at-Risk* of X at the level α is denoted $\mathrm{VaR}_\alpha(X)$ with

$$\mathrm{VaR}_\alpha(X) := \inf\{x \in \mathbb{R} : \mathbb{P}[X \leq x] \geq \alpha\}.$$

Value-at-Risk is a measure of risk and is often used in the financial industry.

1. Sketch a possible graph of the distribution function of X and illustrate on the graph how to find $\mathrm{VaR}_{0.99}(X)$.

A bank models its loss over one day on its investment portfolio by a random variable L. Suppose that $L = L_0(1+i)$, with random variable $(1+i) \sim \log\mathcal{N}(\mu, \sigma^2)$ and constants $\mu, \sigma, L_0 \in (0, \infty)$.

2. Find an expression for $\mathrm{VaR}_\alpha(L)$ in terms of Φ, the standard normal distribution function. Justify every step in your calculation.

3. Interpret the meaning of $\mathrm{VaR}_{0.99}(L)$ in words.

Question 10 The probability density function (pdf) of a random variable $Y \sim \mathcal{N}(\mu, \sigma^2)$ is

$$f_Y(y) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2}\frac{(y-\mu)^2}{\sigma^2}\right), \qquad \forall y \in \mathbb{R}.$$

Let the random variable $X \sim \log\mathcal{N}(\mu, \sigma^2)$.

1. Show that the pdf f_X of X is

$$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}}\frac{1}{x} \exp\left(-\frac{1}{2}\frac{(\ln(x)-\mu)^2}{\sigma^2}\right), \qquad \forall x > 0.$$

2. Show that

$$\mathbb{E}(X^p) = \exp\left(p\mu + \frac{1}{2}p^2\sigma^2\right), \qquad \forall p \in \mathbb{R}.$$

Numerical solutions: Q1.1. $\mathbb{E}(1+i_n) = 1.056667$, $\mathrm{var}(1+i_n) = 0.000355$, $\mathrm{sd}(1+i_n) = 0.0188$; Q1.2.a. $\mathbb{E}(S_3) = 1.179816$, $\mathrm{var}(S_3) = 0.001328$; Q1.2.b. $\mathbb{E}(S_3) = 1.180938$, $\mathrm{var}(S_3) = 0.003889$; Q2. 0.375; Q3. 0.6578; Q4.1. $E[1+i_n] = 1.05$, $E[(1+i_n)^2] = 1.10266$; Q4.2. $E[A_1] = 1.05$, $E[A_2] = 2.1525$; Q4.3. $\mathrm{var}(A_2) = 0.000849$; Q5.1. 15.645487; Q5.2. 281.138066; Q5.3. $\mathbb{E}(A_{10}^3) = 5850.969145$; Q6.1. $\mu = 0.043333$, $\sigma = 0.104476$; Q6.2. 0.10775; Q6.3. 0.25869; Q7.2. mean = 1.592716, sd = 0.340982; Q8.2. 0.14842; Q9.2. $\mathrm{VaR}_\alpha(L) = L_0 e^{\mu+\sigma\Phi^{-1}(\alpha)}$.

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Solutions to Tutorial 4 – Stochastic interest rate models

Solution 1

1. Note: in the following calculation the variance is sensitive with respect to rounding errors in the expectation.

$$\mathbb{E}(1+i_n) = \frac{1}{3} \times 1.03 + \frac{2}{3} \times 1.07 = 1.056667$$

$$\mathbb{E}((1+i_n)^2) = \frac{1}{3} \times 1.03^2 + \frac{2}{3} \times 1.07^2 = 1.1169$$

$$\mathrm{var}(1+i_n) = \mathbb{E}((1+i_n)^2) - (\mathbb{E}(1+i_n))^2 = 1.1169 - (1.056667)^2 = 0.000355$$

$$\mathrm{sd}(1+i_n) = \sqrt{\mathrm{var}(1+i_n)} = 0.000355^{\frac{1}{2}} = 0.0188$$

2. (a) As

$$S_T = \prod_{n=1}^{T} (1+i_n)\,,$$

under the i.i.d. assumption, we have

$$\begin{aligned}
\mathbb{E}(S_3) &= \mathbb{E} \prod_{n=1}^{3} (1+i_n) \\
&= \prod_{n=1}^{3} \mathbb{E}\,(1+i_n) \qquad \text{by independence} \\
&= \prod_{n=1}^{3} 1.056667 \qquad \text{as identically distributed} \\
&= 1.179816
\end{aligned}$$

and

$$\begin{aligned}
\mathbb{E}(S_3^2) &= \mathbb{E} \prod_{n=1}^{3} (1+i_n)^2 \\
&= \prod_{n=1}^{3} \mathbb{E}\left((1+i_n)^2\right) \qquad \text{by independence} \\
&= \prod_{n=1}^{3} 1.1169 \qquad \text{as identically distributed} \\
&= 1.393294
\end{aligned}$$

Hence $\mathrm{var}(S_3) = \mathbb{E}(S_3^2) - (\mathbb{E}(S_3))^2 = 1.393294 - (1.179816)^2 = 0.001328$.

(b) As before, $S_T = \prod_{n=1}^{T}(1+i_n)$. Under the alternative assumption that $i_1 = i_2 = \cdots$ with probability 1, we have

$$\begin{aligned}
\mathbb{E}(S_3) &= \mathbb{E}\prod_{n=1}^{3}(1+i_n) \\
&= \mathbb{E}\prod_{n=1}^{3}(1+i_1) \qquad \text{as } i_1 = i_2 = \cdots \text{ with probability 1} \\
&= \mathbb{E}\left((1+i_1)^3\right) \\
&= \frac{1}{3}\times 1.03^3 + \frac{2}{3}\times 1.07^3 \\
&= 1.180938
\end{aligned}$$

and

$$\begin{aligned}
\mathbb{E}(S_3^2) &= \mathbb{E}\prod_{n=1}^{3}(1+i_n)^2 \\
&= \mathbb{E}\prod_{n=1}^{3}(1+i_1)^2 \qquad \text{as } i_1 = i_2 = \cdots \text{ with probability 1} \\
&= \mathbb{E}\left((1+i_1)^6\right) \\
&= \frac{1}{3}\times 1.03^6 + \frac{2}{3}\times 1.07^6 \\
&= 1.398504
\end{aligned}$$

Hence $\text{var}(S_3) = \mathbb{E}(S_3^2) - (\mathbb{E}(S_3))^2 = 1.398504 - (1.180938)^2 = 0.003889$.

Solution 2

The accumulation of £1 from the start of year 0 to the end of year 2 is given by the random variable

$$A = (1+i_0)(1+i_1)(1+i_2).$$

Considering the possible values of the r.v. A and noting that $1.06^3 = 1.191016 < 1.2$, we need only consider the following possible values of A

Accumulation of £1	Probability of this value
$1.06 \times 1.06 \times 1.08$	$0.5 \times 0.25 = 0.125$
$1.06 \times 1.08 \times 1.06$	$0.25 \times 0.25 = 0.0625$
$1.06 \times 1.08 \times 1.08$	$0.25 \times 0.5 = 0.125$
$1.06 \times 1.08 \times 1.10$	$0.25 \times 0.25 = 0.0625$
TOTAL PROBABILITY	0.375

since all other possible values of A result in an accumulation of less than £1.20. Hence the probability that an investment of £1 at time 0 will accumulate to more than £1.20 at time 3 is 0.375.

Solution 3

Denote the accumulated value at time 12 by A, with $A = 10\,S_{12} = 10\prod_{t=1}^{12}(1+i_t)$.

$$\begin{aligned}
\mathbb{E}[A] &= 10\prod_{t=1}^{12}\mathbb{E}[1+i_t] \qquad \text{by independence} \\
&= 10\times 1.06^{12} \qquad \text{as identically distributed} \\
&= 20.122
\end{aligned}$$

and

$$\begin{aligned}
\mathbb{E}[A^2] &= 100\prod_{t=1}^{12}\mathbb{E}[(1+i_t)^2] \qquad \text{by independence}\\
&= 100\prod_{t=1}^{12}\left(\mathbb{E}[1]+2\mathbb{E}[i_t]+\mathbb{E}[i_t^2]\right)\\
&= 100\prod_{t=1}^{12}\left(\mathbb{E}[1]+2E[i_t]+\mathrm{var}[i_t]+(\mathbb{E}[i_t])^2\right)\\
&= 100\times\left[1+2\times 0.06+0.01^2+0.06^2\right]^{12} \qquad \text{as identically distributed}\\
&= 405.326
\end{aligned}$$

Hence $\mathrm{var}(A) = \mathbb{E}[A^2] - (\mathbb{E}[A])^2 = 405.326 - 20.122^2 = 0.4326$, and $\mathrm{sd}(A) = \sqrt{\mathrm{var}(A)} = \sqrt{0.4326} = 0.6578$.

Hence the standard deviation of the accumulated value at time 12 of an investment of £10 at time 0 is 0.6578.

Solution 4

1. As $(i_n)_n$ are i.i.d. r.v.s, we have

$$E[1+i_n] = E[1+i_1] = 0.2(1.03)+0.6(1.05)+0.2(1.07) = 1.05$$

$$E[(1+i_n)^2] = E[(1+i_1)^2] = 0.2(1.03)^2+0.6(1.05)^2+0.2(1.07)^2 = 1.10266.$$

2. We have $A_1 = 1+i_1$ and thus $E[A_1] = 1.05$. In general

$$E[A_n] = \ddot{s}_{\overline{n}|}$$

calculated at the effective rate of interest $i = E[i_1] = 0.05$. Hence

$$E[A_2] = \ddot{s}_{\overline{2}|} = 2.1525.$$

3. From the recursion $A_n = (1+A_{n-1})(1+i_n)$ we have

$$\begin{aligned}
E[A_2^2] &= E[(1+A_1)^2(1+i_2)^2]\\
&= E[(1+A_1)^2]\times E[(1+i_2)^2], \quad \text{by independence of } i_1, i_2.
\end{aligned}$$

Now

$$\begin{aligned}
E[(1+A_1)^2] &= (1+2E[(1+i_1)]+E[(1+i_1)^2])\\
&= 1+2\times 1.05+1.10266\\
&= 4.20266,
\end{aligned}$$

and so

$$E[(A_2)^2] = 4.20266\times 1.10266 = 4.634105,$$

and thus

$$\mathrm{var}[A_2] = 4.634105-(2.1525)^2 = E[A_2^2]-E[A_2]^2 = 0.000849.$$

Solution 5

1. As $E[A_n] = \ddot{s}_{\overline{n}|}$ at the effective rate of interest $i = E[i_t] = 0.08$,

$$E[A_{10}] = \frac{1.08^{10} - 1}{0.08/1.08} = 15.645487.$$

2. Define $m_p(n) = \mathbb{E}(A_n^p)$, so that $m_1(n) = E[A_n]$, $m_2(n) = \mathbb{E}[A_n^2]$ and $m_3(n) = \mathbb{E}[A_n^3]$. Then

$$\begin{aligned} m_2(n+1) &= \left(1 + 2E[i_t] + E[i_t^2]\right) \times [m_2(n) + 2m_1(n) + 1] \\ &= (1 + 2 \times 0.08 + 0.042) \times [m_2(n) + 2m_1(n) + 1] \\ &= 1.202 \times [m_2(n) + 2m_1(n) + 1]. \end{aligned}$$

We wish to calculate $\mathbb{E}(A_{10}^2) = m_2(10)$. As $m_2(1) = \mathbb{E}(A_1^2) = E[(1 + i_1)^2] = 1.202$ and $m_1(1) = \ddot{s}_{\overline{1}|} = 1.08$, we obtain by recursion the values in Table 1. Hence $\mathbb{E}(A_{10}^2) = 281.138066$.

3. Let $1+k = E[(1+i_t)^3] = 1+3E[i_t]+3E[i_t^2]+E[i_t^3] = 1+3\times 0.08+3\times 0.042+0.013 = 1.379$.

$$\begin{aligned} m_3(n+1) &= E[A_{n+1}^3] \\ &= E[(A_n + 1)^3 (1 + i_{n+1})^3] \\ &= E[(A_n + 1)^3] E[(1 + i_{n+1})^3] \qquad \text{by independence} \\ &= E[1 + 3A_n + 3A_n^2 + A_n^3](1 + k) \\ &= [1 + 3m_1(n) + 3m_2(n) + m_3(n)] \times 1.379 \end{aligned}$$

As $m_3(1) = \mathbb{E}(A_1^3) = E[(1 + i_1)^3] = 1.379$, we obtain by recursion the values in Table 1. Hence $\mathbb{E}(A_{10}^3) = 5850.969145$.

n	$\ddot{s}_{\overline{n}\|} = m_1(n)$	$m_2(n)$	$m_3(n)$
1	1.08000	1.202000	1.379000
2	2.246400	5.243124	12.721275
3	3.506112	12.904581	49.905799
4	4.866601	25.141999	138.090132
5	6.335929	43.121992	315.950871
6	7.922803	68.266208	641.682670
7	9.636628	102.304401	1201.453339
8	11.487558	147.338342	2121.283189
9	13.486563	205.918777	3583.691267
10	15.645488	281.138066	5850.969145

Table 1: Recursion for Question 5

Solution 6

1. $\ln(1+i_t) \sim \mathcal{N}(\mu, \sigma^2)$

$$\begin{aligned}
\mathbb{E}[1+i_t] = e^{\mu+\sigma^2/2} = 1.05 \qquad &\Rightarrow \qquad \mu + \sigma^2/2 = \ln(1.05). \\
\text{var}(1+i_t) &= (e^{\mu+\sigma^2/2})^2(e^{\sigma^2}-1) \\
0.11^2 &= (1.05)^2(e^{\sigma^2}-1) \\
\Rightarrow e^{\sigma^2} &= 1.010975 \\
\Rightarrow \sigma^2 &= 0.010915
\end{aligned}$$

Hence $\mu = 0.043333$ and $\sigma = 0.104476$.

2. Using $\Phi(z)$ to denote the c.d.f. of the standard normal distribution,

$$\begin{aligned}
\mathbb{P}[1.04 < (1+i_t) \leq 1.07] =& \mathbb{P}[(1+i_t) \leq 1.07] - \mathbb{P}[(1+i_t) \leq 1.04] \\
=& \Phi\left(\frac{\ln(1.07)-\mu}{\sigma}\right) - \Phi\left(\frac{\ln(1.04)-\mu}{\sigma}\right) \\
=& \Phi(0.23283) - \Phi(-0.03936) \\
=& \Phi(0.23283) - (1 - \Phi(0.03936))
\end{aligned}$$

Looking up the tables of the CDF of the standard normal distribution, $\Phi(0.23) = 0.59095$ and $\Phi(0.24) = 0.59483$. Hence by linear interpolation,

$$\begin{aligned}
\Phi(0.23283) &= \Phi(0.23) + \frac{0.23283-0.23}{0.24-0.23}(\Phi(0.24) - \Phi(0.23)) \\
&= 0.59095 + \frac{0.23283-0.23}{0.24-0.23}(0.59483 - 0.59095) = 0.59205.
\end{aligned}$$

Similarly, $\Phi(0.03) = 0.51197$ and $\Phi(0.04) = 0.51595$. Hence by linear interpolation,

$$\begin{aligned}
\Phi(0.03936) &= \Phi(0.03) + \frac{0.03936-0.03}{0.04-0.03}(\Phi(0.04) - \Phi(0.03)) \\
&= 0.51197 + \frac{0.03936-0.03}{0.04-0.03}(0.51595 - 0.51197) = 0.51570.
\end{aligned}$$

Hence $\mathbb{P}[1.04 < (1+i_t) \leq 1.07] = 0.59205 - (1 - 0.51570) = 0.10775$.

3. By the i.i.d. assumption, $\ln(S_2) \sim \mathcal{N}(2\mu, 2\sigma^2)$. Then

$$\begin{aligned}
\mathbb{P}[100S_2 > 120] &= 1 - \mathbb{P}[100S_2 \leq 120] \\
&= 1 - \mathbb{P}[\ln(100) + \ln(S_2) \leq \ln(120)] \\
&= 1 - \mathbb{P}[\ln(S_2) \leq \ln(120) - \ln(100)] \\
&= 1 - \mathbb{P}[\frac{\ln(S_2) - 2\mu}{\sqrt{(2\sigma^2)}} \leq \frac{\ln(120) - \ln(100) - 2\mu}{\sqrt{(2\sigma^2)}}] \\
&= 1 - \Phi\left(\frac{\ln(\frac{120}{100}) - 2\mu}{\sqrt{2}\sigma}\right) \\
&= 1 - \Phi(0.64741)
\end{aligned}$$

Looking up the tables of the CDF of the standard normal distribution, $\Phi(0.64) = 0.73891$ and $\Phi(0.65) = 0.74215$. Hence by linear interpolation,

$$\begin{aligned}
\Phi(0.64741) &= \Phi(0.64) + \frac{0.64741-0.64}{0.65-0.64}(\Phi(0.65) - \Phi(0.64)) \\
&= 0.73891 + \frac{0.64741-0.64}{0.65-0.64}(0.74215 - 0.73891) = 0.74131.
\end{aligned}$$

Hence $\mathbb{P}[100S_2 > 120] = 1 - 0.74131 = 0.25869$.

Solution 7

1. For some unknown parameters μ and σ^2, we know

$$\ln(1+i_t) \sim \mathcal{N}(\mu, \sigma^2).$$

We begin by finding μ and σ^2. We are told $\mathbb{E}[1+i_t] = 1.10$ and $\text{sd}(1+i_t) = 0.15$.

$$\mathbb{E}[1+i_t] = e^{\mu+0.5\sigma^2} = 1.1 \qquad \Rightarrow \qquad \mu + 0.5\sigma^2 = \ln(1.1)$$

$$\begin{aligned} \text{var}(1+i_t) = e^{2\mu+\sigma^2}\left(e^{\sigma^2}-1\right) &= (0.15)^2 \\ \Rightarrow \quad e^{\sigma^2} - 1 &= \frac{0.15^2}{1.1^2} \\ \Rightarrow \quad \sigma^2 &= \ln\left(1+\frac{0.15^2}{1.1^2}\right) = 0.018424 \end{aligned}$$

Hence $\mu = 0.086098$ and $\sigma = 0.135735$.

Next, notice that

$$\ln \upsilon_t = \ln((1+i_t)^{-1}) = -\ln(1+i_t),$$

i.e. $\ln \upsilon_t \sim \mathcal{N}(-\mu, \sigma^2)$. Then

$$\mathbb{E}(\upsilon_t) = e^{-\mu+0.5\sigma^2} = e^{-0.086098+0.5\times 0.018424} = 0.925995$$

and

$$\text{var}(\upsilon_t) = e^{-2\mu+\sigma^2}\left(e^{\sigma^2}-1\right) = e^{-2\times 0.086098+0.018424}\left(e^{0.018424}-1\right) = 0.0159444.$$

Hence $\text{sd}(\upsilon_t) = \sqrt{\text{var}(\upsilon_t)} = \sqrt{0.0159444} = 0.126271$.

2. Denote the present value of the payment stream by X. Then

$$X = \upsilon_1\upsilon_2 + \upsilon_1\upsilon_2\upsilon_3\upsilon_4$$

Using the i.i.d. assumption, the mean is

$$\begin{aligned} \mathbb{E}[X] &= \mathbb{E}[\upsilon_1]\mathbb{E}[\upsilon_2] + \mathbb{E}[\upsilon_1]\mathbb{E}[\upsilon_2]\mathbb{E}[\upsilon_3]\mathbb{E}[\upsilon_4] \\ &= 0.925995^2 + 0.925995^4 \\ &= 1.592716 \end{aligned}$$

For the standard deviation, as

$$\mathbb{E}[\upsilon_t^2] = (\text{sd}(\upsilon_t))^2 + \mathbb{E}[\upsilon_t]^2 = 0.126271^2 + 0.925995^2 = 0.873411$$

then

$$\begin{aligned} \mathbb{E}[X^2] &= \mathbb{E}[\upsilon_1^2]\mathbb{E}[\upsilon_2^2]\left(1 + 2\mathbb{E}[\upsilon_3]\mathbb{E}[\upsilon_4] + \mathbb{E}[\upsilon_3^2]\mathbb{E}[\upsilon_4^2]\right) \\ &= 0.873411^2\left(1 + 2\times 0.925995^2 + 0.873411^2\right) \\ &= 2.653013 \end{aligned}$$

Hence

$$\text{var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = 2.653013 - (1.592716)^2 = 0.116269,$$

and thus $\text{sd}(X) = \sqrt{\text{var}(X)} = \sqrt{0.116269} = 0.340982$.

Solution 8

1. Let S_n be the accumulated value after n years of 1 unit invested at time 0 and let i_t be the yield from $t-1$ to t. As

$$S_n = (1+i_1)(1+i_2)\cdots(1+i_n),$$

and $\mathbb{E}[1+i_t] = (1+j)$ then

$$\begin{aligned}\mathbb{E}[S_n] &= \mathbb{E}[(1+i_1)(1+i_2)\cdots(1+i_n)] \\ &= \mathbb{E}[(1+i_1)]E[(1+i_2)]\cdots E[(1+i_n)] \quad \text{as yields are independent} \\ &= (1+j)^n \quad \text{as yields are identically distributed,}\end{aligned}$$

and as $\mathbb{E}[(1+i_t)^2] = \mathbb{E}[1+2i_t+i_t^2] = 1+2j+(j^2+s^2)$, we get

$$\begin{aligned}\mathbb{E}[S_n^2] &= \mathbb{E}[((1+i_1)(1+i_2)\cdots(1+i_n))^2] \\ &= \mathbb{E}[(1+i_1)^2]E[(1+i_2)^2]\cdots E[(1+i_n)^2] \quad \text{as yields are independent} \\ &= \left(1+2j+(j^2+s^2)\right)^n \quad \text{as yields are identically distributed}\end{aligned}$$

Hence

$$\begin{aligned}\operatorname{var}(S_n) = \mathbb{E}[S_n^2] - (\mathbb{E}[S_n])^2 &= (1+2j+(j^2+s^2))^n - (1+j)^{2n} \\ &= ((1+j)^2+s^2)^n - (1+j)^{2n}\end{aligned}$$

2. We are told $i_1, i_2, \ldots$ are i.i.d. with $\ln(1+i_n) \sim \mathcal{N}(0.08, 0.04^2)$. Defining $S_n = (1+i_1)\cdots(1+i_n)$, we find

$$\ln(S_n) = \ln\prod_{k=1}^{n}(1+i_k) = \sum_{k=1}^{n}\ln(1+i_k).$$

By the properties of the lognormal distribution, this means

$$\ln(S_n) \sim \mathcal{N}(0.08n, 0.04^2 n).$$

We wish to find $\mathbb{P}[1000S_{16} > 4250]$, i.e.

$$\begin{aligned}\mathbb{P}[1000S_{16} > 4250] &= \mathbb{P}[S_{16} > 4.250] \\ &= 1 - \mathbb{P}[S_{16} \le 4.250] \\ &= 1 - \mathbb{P}[\ln(S_{16}) \le \ln(4.250)] \\ &= 1 - \mathbb{P}\left[\frac{\ln(S_{16}) - 0.08\times 16}{\sqrt{0.04^2\times 16}} \le \frac{\ln(4.250) - 0.08\times 16}{\sqrt{0.04^2\times 16}}\right] \\ &= 1 - \Phi\left(\frac{\ln(4.250) - 0.08\times 16}{\sqrt{0.04^2\times 16}}\right) \\ &= 1 - \Phi(1.04324).\end{aligned}$$

Looking up the tables of the CDF of the standard normal distribution, $\Phi(1.04) = 0.85083$ and $\Phi(1.05) = 0.85314$. Hence by linear interpolation,

$$\begin{aligned}\Phi(1.04324) &= \Phi(1.04) + \frac{1.04324 - 1.04}{1.05 - 1.04}(\Phi(1.05) - \Phi(1.04)) \\ &= 0.85083 + \frac{1.04324 - 1.04}{1.05 - 1.04}(0.85314 - 0.85083) = 0.85158.\end{aligned}$$

Thus $\mathbb{P}[1000S_{16} > 4250] = 1 - 0.85158 = 0.14842$.

Solution 9

1. See other file for possible sketches.

2. As $L_0 \in (0, \infty)$ then $\ln(L_0) \in (-\infty, \infty)$. By using the properties of a normally distributed random variable, we can show

$$\ln(L) \sim \mathcal{N}(\ln(L_0) + \mu, \sigma^2),$$

with both parameters finite. Next,

$$\begin{aligned}
\text{VaR}_\alpha(L) &= \inf\{x \in \mathbb{R} : \mathbb{P}\left[L \leq x\right] \geq \alpha\} \\
&= \inf\{x \in \mathbb{R} : \mathbb{P}\left[\frac{\ln(L) - \ln(L_0) - \mu}{\sigma} \leq \frac{\ln(x) - \ln(L_0) - \mu}{\sigma}\right] \geq \alpha\} \\
&= \inf\{x \in \mathbb{R} : \Phi\left(\frac{\ln(x) - \ln(L_0) - \mu}{\sigma}\right) \geq \alpha\}.
\end{aligned}$$

As $\Phi(z) \geq y \Leftrightarrow z \geq \Phi^{-1}(y)$, we get

$$\begin{aligned}
\text{VaR}_\alpha(L) &= \inf\{x \in \mathbb{R} : \left(\frac{\ln(x) - \ln(L_0) - \mu}{\sigma}\right) \geq \Phi^{-1}(\alpha)\} \\
&= \inf\{x \in \mathbb{R} : \ln\left(\frac{x}{L_0}\right) \geq \mu + \sigma\Phi^{-1}(\alpha)\}.
\end{aligned}$$

We recall that the exponential function is monotonically increasing so that if we apply the exponential function to both sides of the inequality, then we know that the inequality is preserved, i.e.

$$\begin{aligned}
\text{VaR}_\alpha(L) &= \inf\{x \in \mathbb{R} : \frac{x}{L_0} \geq \exp\left(\mu + \sigma\Phi^{-1}(\alpha)\right)\} \\
&= \inf\{x \in \mathbb{R} : x \geq L_0 \exp\left(\mu + \sigma\Phi^{-1}(\alpha)\right)\} \\
&= L_0 \exp\left(\mu + \sigma\Phi^{-1}(\alpha)\right).
\end{aligned}$$

3. There is a 99% probability that the loss on the bank's investment portfolio over one day is less than $\text{VaR}_{0.99}(L)$.

   Alternatively, there is a 1% probability that the loss on the bank's investment portfolio over one day is greater than $\text{VaR}_{0.99}(L)$.

Solution 10

1. Recall $X \sim \log\mathcal{N}(\mu, \sigma^2) \Leftrightarrow \ln(X) \sim N(\mu, \sigma^2)$. For $x > 0$,

$$\mathbb{P}[X \leq x] = \mathbb{P}[\ln(X) \leq \ln(x)] = \int_{-\infty}^{\ln(x)} f_Y(y)dy.$$

Change of variable: set $y = \ln(z)$. Then as $dy = 1/zdz$,

$$\mathbb{P}[X \leq x] = \int_0^x \frac{1}{z} f_Y(\ln(z))dz = \int_0^x \frac{1}{z}\frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{1}{2}\frac{(\ln(z) - \mu)^2}{\sigma^2}\right) dz.$$

Hence $f_X(x) = \frac{1}{\sigma\sqrt{2\pi}}\frac{1}{x} \exp\left(-\frac{1}{2}\frac{(\ln(x)-\mu)^2}{\sigma^2}\right)$, for all $x > 0$.

2. Show that

$$\mathbb{E}\left(X^p\right) = \exp\left(p\mu + \frac{1}{2}p^2\sigma^2\right), \qquad \forall p \in \mathbb{R}.$$

Using the result in Solution 6.1,

$$\begin{aligned}\mathbb{E}\left(X^p\right) &= \int_0^\infty x^p f_X(x)dx \\ &= \int_0^\infty x^{p-1}\frac{1}{\sigma\sqrt{2\pi}}\exp\left(-\frac{1}{2}\frac{(\ln(x)-\mu)^2}{\sigma^2}\right)dx.\end{aligned}$$

Change of variable: set $x = e^y$. Then as $dx = e^y dy$,

$$\begin{aligned}\mathbb{E}\left(X^p\right) &= \int_{-\infty}^\infty e^{y(p-1)}\frac{1}{\sigma\sqrt{2\pi}}\exp\left(-\frac{1}{2}\frac{(y-\mu)^2}{\sigma^2}\right)e^y dy \\ &= \frac{1}{\sigma\sqrt{2\pi}}\int_{-\infty}^\infty \exp\left(py - \frac{1}{2}\frac{(y-\mu)^2}{\sigma^2}\right)dy.\end{aligned}$$

Completing the square in the exponent of the exponential,

$$\mathbb{E}\left(X^p\right) = \frac{1}{\sigma\sqrt{2\pi}}\exp\left(p\mu + \frac{1}{2}p^2\sigma^2\right)\int_{-\infty}^\infty \exp\left(-\frac{1}{2}\frac{(y-(\mu+\sigma^2 p))^2}{\sigma^2}\right)dy.$$

Change of variable: set $z = \frac{y-(\mu+\sigma^2 p)}{\sigma}$. Then as $dy = \sigma dz$ and using ϕ to denote the pdf of the standard normal distribution,

$$\begin{aligned}\mathbb{E}\left(X^p\right) &= \frac{1}{\sigma\sqrt{2\pi}}\exp\left(p\mu + \frac{1}{2}p^2\sigma^2\right)\int_{-\infty}^\infty \exp\left(-\frac{1}{2}z^2\right)\sigma dz \\ &= \exp\left(p\mu + \frac{1}{2}p^2\sigma^2\right)\int_{-\infty}^\infty \frac{1}{\sqrt{2\pi}}\exp\left(-\frac{1}{2}z^2\right)dz \\ &= \exp\left(p\mu + \frac{1}{2}p^2\sigma^2\right)\int_{-\infty}^\infty \phi(z)dz \\ &= \exp\left(p\mu + \frac{1}{2}p^2\sigma^2\right).\end{aligned}$$